Junhyun An

Statistics & Mathematics, Seoul National University | anjunhyun@snu.ac.kr | +82-10-2415-7065

https://anjunhyun.github.io/anjunhyun/

Education

Seoul National University, BS in Statistics and Mathematical Sciences — Mar 2021 – Present
Korea Science Academy of KAIST (KSA) — 2018 – 2020

Research Interests

Knot Theory • Combinatorial Topology • Graph Theory • Low-Dimensional Topology

Research Experience

Research & Education (R&E) — May 2019 – Dec 2019
Advisor: Prof. Gyotaek Jin

The Smallest Lattice Cube Containing Nontrivial Band Knots

- Proved that every band knot in a $3 \times 3 \times 3$ lattice cube is ambient isotopic to the unknot.
- Shown existence of a Möbius strip in $2 \times 2 \times 2$ and a trefoil knot in $4 \times 4 \times 4$ by computational enumeration.

Independent Research — May 2020 – Dec 2020, Feb 2023 – Sep 2023

Computational Complexity and Approximation of the Rural Postman Problem in Euclidean Graphs

- Proved NP-hardness by showing that Euclidean TSP is a special case.
- Developed a polynomial-time upper bound using minimum matching and a minimum spanning tree.

Undergraduate Research Internship — Dec 2024 – Dec 2025
Advisor: Prof. Jisu Kim

Periodicity Testing for Time Series

- Co-developed a periodicity-testing method using one-dimensional persistent homology of time-delay embeddings and proved asymptotically valid confidence bounds for the resulting features under the paper's assumptions.

Random Graph Models for Knot Generation

- Developed computational methods to study knot distributions in random graph embeddings.

Persistent Homology of Subsampled Paley Graphs

- Observed relationships between persistent homology and Gauss sum magnitudes, including differences between prime and prime-power fields.

Independent Research — Aug 2026 – Present

Rank-Weighted Coloring Polynomial and the Partial-Dual Genus Polynomial

- Introduced a construction that recovers several existing objects and proved a structural identity through a case analysis tracking preserved quantities.
- Developed the project from definition to manuscript, distinguishing consequences of an existing framework from claims requiring a direct argument.

Academic Activities

Knot Theory Seminar, Organizer and Presenter — Mar 2026 – Jun 2026

- Designed and coordinated a nine-lecture seminar for undergraduate and graduate students, working with two other presenters.
- Delivered several lectures, reviewed the other presentations, and designed all exercises; led solution sessions or provided written solutions.
- Covered knot groups; Alexander, Jones, and Khovanov invariants; braids, virtual knots, and 4D slice genus.

Preprint

[1] D. Park, **J. An**, T. Kim, and J. Kim. "Subsampling Confidence Bound for Persistent Diagram via Time-delay Embedding" *arXiv:2512.06324*, 2025; revised 2026. Co-first author.